Uncertainty-Weighted Test-Time Augmentation for Robust Medical Image Classification

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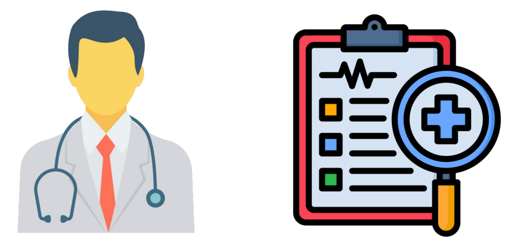
Background and Problem Statement

- Deep learning classifiers for **medical imaging** are routinely deployed with a single forward pass, but these predictions are frequently **overconfident** and **poorly calibrated**.
- **Calibration** is a safety-critical requirement in clinical medicine to prevent systems from creating a dangerous illusion of certainty.
- Poor calibration is compounded by modern cross-entropy training objectives, distribution shifts between scanners, and class-imbalanced medical datasets.
- **Test-Time Augmentation (TTA)** improves robustness, but standard TTA assumes uniform weights, which is fundamentally flawed for medical imaging.



88.1% → 56.5%

PathMNIST accuracy collapse with 50 equal-weight TTA views (*Medeiros D. N (2026) - prior work)*

a 31.6 pp degradation

Research Gaps

Accuracy Degradation
Naive equal-weight TTA frequently degrades accuracy because natural-image transformations can be clinically meaningless or impossible

Calibration Impact
The impact of TTA on predictive calibration has not been systematically studied.

Uncertainty Weighting
It is unknown whether uncertainty-based weighting can recover reliable confidence estimates across diverse modalities.

Mechanism Clarity
The specific conditions under which uncertainty-based weighting actually helps or hurts remain poorly understood

Key Contributions

Systematic Evaluation
Evaluated nine calibration-focused TTA fusion strategies across six distinct medical imaging modalities and three backbones.

Two-factor Explanation
Proposed a two-factor explanation demonstrating that calibration gains depend on entropy informativeness and augmentation safety.

Visualizing Confidence
Introduced "Augmentation Confidence Strips" to expose which augmented views a model trusts or rejects

Practitioner Guide
Distilled empirical findings into an actionable selection guide for applying TTA under calibration constraints.

Methodology & System Pipeline

One trained model → N augmented views → confidence-weighted fusion → calibrated prediction

1. **Train** — ResNet-18 / EffNet-B0, DeiT-Tiny, ImageNet-pretrained, per-dataset head
2. **Augment** — Generate N=10 views per test image (10 augmentation types)
3. **Weight & Fuse** — Per-view softmax → confidence weight w_i → weighted average
4. **Calibrate** — Temperature T fit on validation (LBFGS on NLL)
5. **Evaluate** — Accuracy · AUC · ECE (10-bin) · NLL · inference time

Standard TTA (baseline)
$\bar{p} = 1/N \cdot \Sigma_i p_i$

vs

Uncertainty-weighted (ours)
$\bar{p} = \Sigma_i w_i p_i / \Sigma_i w_i$

Fusion Strategies

Eight soft-weighting strategies below; a Top-K entropy-filter family (top3 / top5 / top7) is also reported as an ablation

Strategy	Description	Weight
Baseline TTA	Every view weighted equally	$w_i = \frac{1}{N}$
Variance-inv	Literal proposal formula → negative finding	$w_i = \frac{1}{var + \varepsilon}$
Max-Prob	Confident views dominate	$w_i = \max(p_i)$
MC-Dropout	Epistemic proxy, no augmentation	*T stochastic passes*
Entropy	Low-entropy views weigh more usual winner	$w_i = e^{-H(p_i)}$
TS-only	Temperature scaling, single pass which is nearly free	*softmax(z / T)*
Variance	Sharp views up (confidence-aligned)	$w_i = \mathrm{var}(p_i)$
TS + Entropy	Best, most consistent calibration	*entropy on z / T*

Used Datasets

Dataset	Modality	Task	Classes	Train size
PathMNIST	Colon pathology (histology)	Multi-class	9	~90,000
DermaMNIST	Dermatoscope (skin)	Multi-class	7	~7,000
PneumoniaMNIST	Chest X-ray	Binary	2	~4,700
BreastMNIST	Breast ultrasound	Binary	2	546
BloodMNIST	Blood-cell microscopy	Multi-class	8	~12,000
OrganAMNIST	Abdominal CT	Multi-class	11	~35,000

All from MedMNIST v2 · 64×64 · pip-installable & pre-split · spanning 5 imaging modalities, binary + multi-class tasks, and a wide size range (546 → ~90k) for a diverse benchmark.

Teck Stack

LIBRARIES: MedMNIST · PyTorch · NumPy · git · pandas

MODELS · ImageNet-pretrained: ResNet-18 · EfficientNet-B0

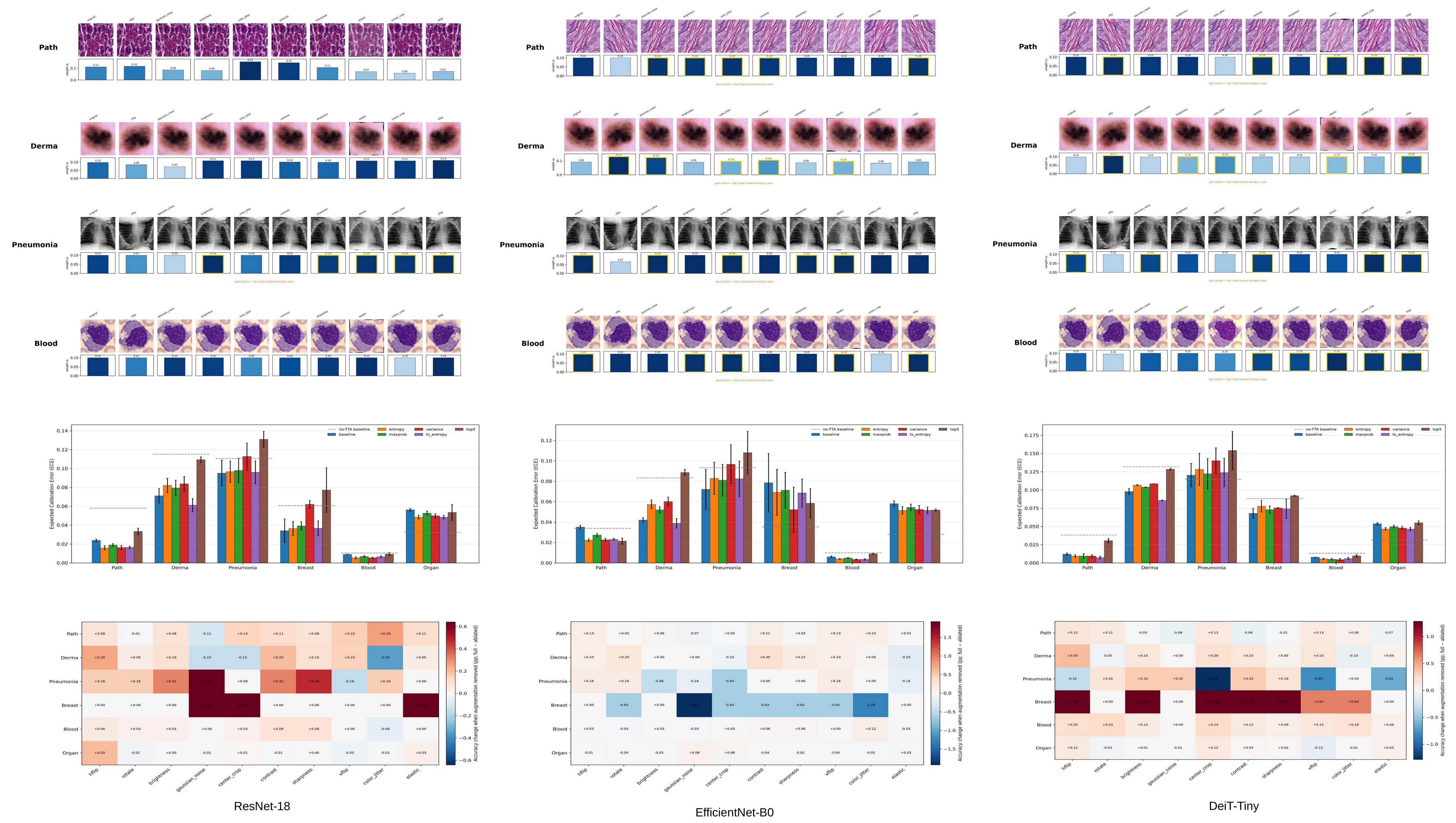
Results

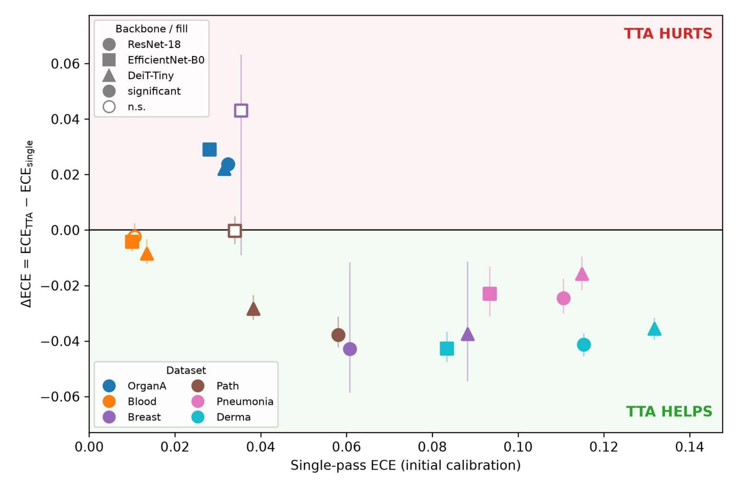
Calibration Improved, Accuracy Maintained

+ 0.04 p
mean accuracy change (Wilcoxon, 6 datasets)

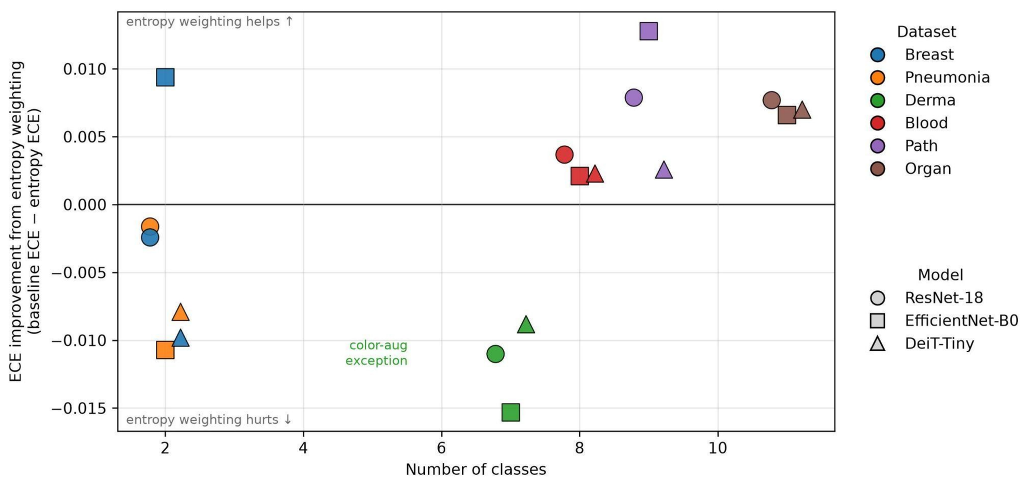
p = 0.165
Wilcoxon signed-rank not significant

≤ 2 / 6
datasets with a significant McNemar accuracy gain





Best uncertainty-aware strategy vs equal-weight TTA (3-seed mean) representative subset of the 11 strategies; dashed = no-TTA baseline



Entropy weighting helps more as classes grow. Binary tasks (Breast, Pneumonia) gain little; Derma is the colour-aug exception (below the line).

−33%
ECE · PathMNIST

−41%
ECE · BloodMNIST

−19%
ECE · DermaMNIST

Conclusions and Recommendation

Calibration, not Accuracy
Uncertainty-weighted TTA leaves accuracy unchanged (a deliberate, tested null) while improving ECE on the multiclass modalities most with TS + Entropy.

A clear recommendation
TS + Entropy consistently offers the best calibration across multiple datasets and backbones. Soft weighting is superior to hard Top-K filtering, and the inverse variance formula should be avoided

Honest, reproducible science
Results fully validated across 5 environments using locked 3-seed evaluation, automated unit tests, and identical evaluation metrics, ensuring our findings are robust and trustworthy.

Why it matters for diagnostics: Trustworthy confidence enables safe triage so confident cases handled automatically, uncertain ones flagged for a clinician at near-zero added cost with TS-only.

Future Directions

- Larger backbones & real hospital data (beyond 64×64)
- Train the model to automatically score the quality of each augmented view
- Edge / streaming deployment TS-only is the cheap path
- Add mathematical guarantees to our confidence scores for safer clinical use